

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

To solve this problem , we need to calculate the quantity of heat required to heat a liter of water from 30 °C to 60 °C , and the quantity of heat required to heat a 1 kg block of aluminum from 30 °C to 60 °C .

For water , the specific heat capacity is given as 1 cal/g °C . We need to calculate the heat required to heat 1000 grams of water (1 liter) from 30 °C to 60 °C . The formula to calculate heat is $Q = mc\Delta T$, where Q is the heat , m is the mass , c is the specific heat capacity , and ΔT is the change in temperature .

$Q_{\text{water}} = 1000 \text{ g} * 1 \text{ cal/g °C} * (60 \text{ °C} - 30 \text{ °C}) = 1000 * 1 * 30 = 30000 \text{ cal}$. Converting this to kilocalories , $Q_{\text{water}} = 30 \text{ Kcal}$.

For aluminum , the specific heat capacity is given as 0 . 215 cal/g °C . We need to calculate the heat required to heat 1000 grams of aluminum (1 kg) from 30 °C to 60 °C . Using the same formula , $Q_{\text{aluminum}} = 1000 \text{ g} * 0 . 215 \text{ cal/g °C} * (60 \text{ °C} - 30 \text{ °C}) = 1000 * 0 . 215 * 30 = 6450 \text{ cal}$. Converting this to kilocalories , $Q_{\text{aluminum}} = 6 . 45 \text{ Kcal}$.

The correct answer is G . 30 Kcal , 6 . 45 Kcal . </think><answer> G </answer>

